ZCAS University

CIT4251 – STRATEGIC IT MANAGEMENT

DEFERRED FINAL EXAMINATION

FRIDAY 19th JUNE 2026

09.00 – 12.00 HOURS

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: Question One in Section A is **compulsory**.
2. Sections B: Answer Three (3) questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so

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SECTION A: QUESTION 1 IS COMPULSORY AND MUST BE ATTEMPTED

Question 1:

Scenario: Read the scenario in **Appendix A** below (also sent to you earlier), and then answer the following questions:

a) Explain how information systems can provide competitive advantage to organizations. (10 Marks)

b) Discuss Porter's three generic strategies and how ZTRL applied them. (10 Marks)

c) Information Systems can be used for three things in an organisation, for automation, for Organizational Learning and for Supporting Strategy. Discuss these in relation to ZamTech Retail and Logistics (ZTRL). (10 Marks)

d) Many companies like ZTRL are continually looking for innovative ways to use information systems for competitive advantage. Discuss five types of innovations that ZTRL could use inn their innovation drive. (10 Marks)

[Total: 40 Marks]

SECTION B: ATTEMPT ONLY THREE QUESTIONS FROM THIS SECTION

Question 2:

a) In order to be successful in cyberspace business using the Business-to-Consumer approach, an organisation should design and build a websites that meets online customers' needs. Discuss the Online Consumers' Hierarchy of Needs in relation to web design. Use a diagram to illustrate this. (10 Marks)

b) Discuss the benefits and challenges of mobile commerce. (10 Marks)

c) Many organisations in cyberspace B2B, B2C, C2C and others sometimes experience shopping cart abandonment, ultimately losing business. Discuss factors contributing to shopping cart abandonment. (5 Marks)

[Total: 25 Marks]

Question 3:

a) Discuss the risks associated with organizational use of social media. (10 Marks)

b) Differentiate between internal and external social media applications. (10 Marks)

c) Explain how organizations like ZTRL can manage negative social media reviews. (5 Marks)

[Total: 25 Marks]

Question 4:

a) Explain the challenges ZTRL face in implementing machine learning techniques for personalized advertising and customer segmentation. (10 Marks)

b) Big Data can be useful to ZTRL. Discuss the characteristics of Big Data. (10 Marks)

c) Evaluate the role of data visualization in decision-making. (5 Marks)

[Total: 25 Marks]

Question 5:

a) Explain the concept of "Productivity Paradox" in relation to technology investments. (10 Marks)

b) The acquisition of software licences may entail consideration of "software rights" depending on the types of software. Discuss the various types of rights in relation to the types of software. (10 Marks)

c) In the event that ZTRL decides to outsource, give reasons for such a decision. (5 Marks)

[Total: 25 Marks]

APPENDIX A

Case Study_Questions_Answers: Digital Transformation at ZamTech Retail and Logistics (ZTRL)

Background

ZamTech Retail and Logistics (ZTRL) is a rapidly growing Zambian retail and distribution company operating supermarkets, online retail platforms, and logistics services across Southern Africa. The company serves over 500,000 customers annually through physical stores, a mobile application, and a web-based e-commerce platform.

Over the past five years, increased competition from international online retailers, changing customer expectations, and rapid digital transformation forced ZTRL to rethink its strategic direction. Customers increasingly demanded online ordering, digital payment systems, real-time delivery tracking, and personalized shopping experiences.

To remain competitive, ZTRL's management initiated a comprehensive Strategic Information Systems (SIS) programme aimed at improving operational efficiency, enhancing customer experience, and increasing profitability.

Strategic Initiatives Undertaken by ZTRL

1. Gaining Competitive Advantage Through Information Systems

ZTRL implemented several information systems to gain competitive advantage:

- An Enterprise Resource Planning (ERP) system integrated procurement, warehousing, finance, and sales operations.
- A Customer Relationship Management (CRM) system enabled customer profiling and personalized marketing campaigns.
- Supply chain systems improved supplier coordination and inventory visibility.
- Mobile payment integration allowed customers to transact using digital wallets and online banking platforms.

The company adopted Porter's competitive strategies by:

- Reducing operational costs through automation,
- Differentiating services using faster delivery and personalized recommendations,
- Building customer intimacy through loyalty programmes and data-driven engagement.

2. Business-to-Consumer (B2C) Electronic Commerce

ZTRL launched a B2C e-commerce platform allowing customers to:

- Browse products online,
- Make electronic payments,

- Track orders in real time,
- Access customer support services.

The company introduced:

- Secure payment gateways,
- Mobile commerce applications,
- Online promotions and discount systems,
- AI-based recommendation engines.

Despite growth in online sales, the company experienced challenges including:

- Cybersecurity threats,
- Website downtime during peak periods,
- Cart abandonment,
- Logistics and delivery delays.

3. Organizational Communication and Collaboration Using Social Media

ZTRL integrated social media into its business strategy using platforms such as:

- Facebook,
- WhatsApp,
- Instagram,
- X (formerly Twitter),
- LinkedIn.

Social media was used for:

- Customer engagement,
- Brand promotion,
- Customer feedback collection,
- Crisis communication,
- Employee collaboration.

Internal collaboration tools such as Microsoft Teams and Slack improved communication between geographically distributed employees.

However, management faced concerns regarding:

- Fake news,
- Negative online reviews,
- Data privacy issues,
- Employee misuse of social media.

4. Business Intelligence, Big Data and Analytics

ZTRL implemented a Business Intelligence (BI) system to analyze large volumes of customer and operational data.

The company collected data from:

- Online transactions,
- Social media interactions,
- Mobile applications,
- Loyalty programmes,
- Sensors in warehouses.

Using analytics dashboards and predictive models, management could:

- Forecast product demand,
- Identify customer buying trends,
- Optimize inventory management,
- Detect fraudulent transactions.

The company also explored machine learning techniques for personalized advertising and customer segmentation.

Challenges included:

- Data quality management,
- Shortage of data science skills,
- Privacy and ethical concerns,
- High infrastructure costs.

5. Developing and Acquiring Information Systems

To support digital transformation, ZTRL evaluated different approaches for acquiring information systems:

- In-house software development,
- Outsourcing,
- Cloud-based Software-as-a-Service (SaaS),
- Off-the-shelf enterprise applications.

The systems development process involved:

1. Systems planning,
2. Requirements analysis,
3. Systems design,
4. Development,
5. Testing,
6. Implementation,

7. Maintenance.

Agile development methodologies were adopted to improve flexibility and faster deployment.

Management also considered:

- Feasibility studies,
- Change management,
- User training,
- Risk management,
- Vendor evaluation.

Although the transformation improved performance, resistance to change among employees and integration challenges between legacy and new systems remained significant concerns.

END OF CASE STUDY